Supplemental Material for "Observer-induced irreversibility in trailing-window early-warning states"

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Tables S1–S12 are generated by `reproducibility_pre/scripts/tab_p1.py` from the CSV files named in their captions; Table S12 lists the files behind the figures and tables of the main text. Seeds and SHA-256 manifests are listed in `reproducibility_pre/README_P1.md`.

Table S1: Data checks (`E1b_splice_report.csv`, `E1d_archive_check.csv`, `E1e_repair_log.csv`). The analysis file is the vega-datasets snapshot to 17 April 2020 continued with the Yahoo download; the data gate (`E1c_data_sanity.py`) rejects non-positive prices and volumes. The one patched volume is the value reported by Tonghuashun (3739.16×10^4 lots of 100 shares); replacing it by the mean of the adjacent days changes no fitted quantity by more than 1.4×10^{-5} (`E4g_volume_sensitivity.csv`).

quantity	value
common days with the independent archive	4774 (2000-01-03 to 2018-12-21)
dates present in only one file	0 / 0
largest relative close difference	5.7e-08
volume ratio, median [min, max]	1.00000 [1.00000, 1.00076]
vega-datasets snapshot vs Yahoo download: common days (2019-01-02 to 2020-04-17)	326
close max. rel. diff.; volume ratio median [min, max]; days off $> 1\%$	1.8e-10; 0.9975 [0.8567, 1.0040]; 27
Yahoo download vs second copy of the Yahoo series: common days (2020-04-01 to 2026-10-16)	1645
close max. rel. diff.; volume ratio median [min, max]; days off $> 1\%$	5.7e-08; 1.0000 [1.0000, 1.0000]; 0
patched days in the 2000–2026 file	2023-05-24: volume 0 → 3,739,160,000 (second vendor)

Table S2: Proposition 1 with linear windows of r_t and r_t^2 on S&P 500 data: antisymmetry measured on the window series and computed from the empirical cross-covariances, with its observer and physical terms, for the data and the time-reversed data (`E4c_linear_decomposition.csv`).

input	pair	measured	theory	O	P
forward	window: mean r^2 (5d) vs mean r^2 (20d)	−0.0282	−0.0283	−0.0283	0.0000
forward	leverage: sum r (20d) vs mean r^2 (20d)	+0.0315	+0.0315	0.0000	+0.0315
forward	mixed: sum r (5d) vs mean r^2 (20d)	+0.0235	+0.0235	+0.0096	+0.0139
reversed	window: mean r^2 (5d) vs mean r^2 (20d)	−0.0282	−0.0283	−0.0283	0.0000
reversed	leverage: sum r (20d) vs mean r^2 (20d)	−0.0315	−0.0315	0.0000	−0.0315
reversed	mixed: sum r (5d) vs mean r^2 (20d)	−0.0042	−0.0042	+0.0096	−0.0139

Table S3: Gaussian observer, closed forms against simulation. Top: $C(s)$ for $\phi = 0.98$, windows (5,20), ten 10^6-step simulations. Bottom: flux-to-drift ratio R, entropy production and Markov divergence of the fitted OU model, closed form against 3×10^5-step simulations, and plug-in symbolic divergences (`T2_cov_check.csv`, `S2_gaussian_irrev.csv`; script `T2_gaussian_observer.py`). The Monte Carlo loss of the exact Bayes observer agrees with Eq. (12) of the main text to within 1.5% of $\mathrm{Var}\, p_{\mathrm{op}}$ for $\phi \leq 0.98$; at $\phi = 0.99$ it is lower by up to 9% because a path holds only about 10^3 independent samples (`S1_gaussian_leak.csv`).

Lag	ij	closed form	simulation	s.e.	rel. err. (%)
0	00	0.968476	0.967764	0.002142	0.074
0	01	0.857512	0.857021	0.002082	0.057
0	10	0.857512	0.857021	0.002082	0.057
0	11	0.878198	0.877668	0.002097	0.060
1	00	0.964633	0.963926	0.002142	0.073
1	01	0.844243	0.843773	0.002073	0.056
1	10	0.869110	0.868599	0.002090	0.059
1	11	0.877367	0.876837	0.002097	0.060
5	00	0.904659	0.904073	0.002127	0.065
5	01	0.782431	0.782042	0.002039	0.050
5	10	0.898883	0.898314	0.002115	0.063
5	11	0.859427	0.858922	0.002086	0.059

ϕ	windows	(w_1, w_2)	R closed/sim	$\dot{e}_p$ closed/sim	d_1 closed/sim	KLD_3	KLD_5
0.9	trailing	(1,4)	1.5636 / 1.5646	6.990 / 6.997	7.602 / 7.607	0.364	0.732
0.9	trailing	(2,8)	1.3111 / 1.3108	4.209 / 4.219	4.670 / 4.683	0.267	0.546
0.9	trailing	(5,20)	1.1319 / 1.1314	2.468 / 2.478	2.793 / 2.806	0.141	0.285
0.9	trailing	(10,40)	1.0760 / 1.0760	1.402 / 1.400	1.554 / 1.552	0.057	0.120
0.9	trailing	(20,80)	1.0513 / 1.0507	0.672 / 0.682	0.716 / 0.727	0.020	0.041
0.9	trailing	(40,160)	1.0403 / 1.0390	0.287 / 0.292	0.296 / 0.302	0.006	0.012
0.9	current+20	(1,20)	0.8469 / 0.8473	3.467 / 3.457	3.591 / 3.580	0.144	0.264
0.95	trailing	(1,4)	1.6772 / 1.6788	7.906 / 7.890	8.276 / 8.254	0.299	0.408
0.95	trailing	(2,8)	1.3371 / 1.3368	5.179 / 5.204	5.524 / 5.548	0.193	0.335
0.95	trailing	(5,20)	1.1306 / 1.1308	3.644 / 3.635	4.043 / 4.027	0.106	0.241
0.95	trailing	(10,40)	1.0698 / 1.0701	2.490 / 2.486	2.810 / 2.801	0.067	0.136
0.95	trailing	(20,80)	1.0412 / 1.0413	1.434 / 1.453	1.590 / 1.605	0.030	0.059
0.95	trailing	(40,160)	1.0282 / 1.0285	0.692 / 0.708	0.739 / 0.754	0.011	0.022
0.95	current+20	(1,20)	1.0866 / 1.0878	4.486 / 4.487	4.614 / 4.612	0.128	0.182
0.98	trailing	(1,4)	1.7499 / 1.7495	8.538 / 8.512	8.704 / 8.675	0.204	0.356
0.98	trailing	(2,8)	1.3446 / 1.3475	5.937 / 5.902	6.115 / 6.075	0.133	0.216
0.98	trailing	(5,20)	1.1238 / 1.1251	4.848 / 4.796	5.141 / 5.078	0.075	0.105
0.98	trailing	(10,40)	1.0625 / 1.0620	3.964 / 4.011	4.342 / 4.394	0.043	0.090
0.98	trailing	(20,80)	1.0338 / 1.0328	2.877 / 2.969	3.235 / 3.343	0.032	0.063
0.98	trailing	(40,160)	1.0197 / 1.0191	1.769 / 1.823	1.980 / 2.046	0.016	0.031
0.98	current+20	(1,20)	1.2951 / 1.2969	5.449 / 5.438	5.535 / 5.522	0.064	0.106
0.99	trailing	(1,4)	1.7748 / 1.7713	8.765 / 8.835	8.851 / 8.923	0.147	0.270
0.99	trailing	(2,8)	1.3456 / 1.3421	6.229 / 6.292	6.326 / 6.392	0.102	0.185
0.99	trailing	(5,20)	1.1200 / 1.1186	5.404 / 5.439	5.587 / 5.634	0.058	0.097
0.99	trailing	(10,40)	1.0587 / 1.0576	4.807 / 4.837	5.094 / 5.146	0.037	0.056
0.99	trailing	(20,80)	1.0306 / 1.0295	3.962 / 4.064	4.338 / 4.477	0.022	0.053
0.99	trailing	(40,160)	1.0168 / 1.0161	2.884 / 2.901	3.242 / 3.262	0.014	0.029
0.99	current+20	(1,20)	1.3789 / 1.3781	5.861 / 5.924	5.913 / 5.978	0.053	0.070

Table S4: Factorization $\dot{e}_p = 2\omega_{\mathrm{fit}}^2 \mathcal{G}$ (Proposition 2) for the Gaussian observer, $\phi = 0.98$, windows $(w, 4w)$, with observation noise η^2 relative to the window variance; raw (unstandardized) coordinates (`T2b_ep_factorization.csv`).

w	η^2	$\dot{e}_p$	ω_{fit}	$\mathcal{G}$	$d_{\min}$	residual
5	0	4.848	−0.0117	1.764e+04	0.000567	4e-15
5	0.001	1.217	−0.0120	4240	0.00228	-7e-16
5	0.01	0.2099	−0.0123	698.5	0.0124	6e-17
5	0.1	0.01499	−0.0118	53.48	0.101	3e-18
10	0	3.964	−0.0094	2.246e+04	0.000312	9e-16
10	0.001	0.6784	−0.0099	3481	0.00189	-3e-16
10	0.01	0.1049	−0.0102	507.7	0.0118	6e-17
10	0.1	0.008391	−0.0098	43.62	0.103	2e-18
20	0	2.877	−0.0064	3.485e+04	0.000176	-4e-16
20	0.001	0.3304	−0.0070	3393	0.00165	6e-16
20	0.01	0.04486	−0.0072	438.4	0.0118	1e-17
20	0.1	0.003836	−0.0069	40.41	0.107	-3e-18
49	0	1.481	−0.0028	9.666e+04	8.23e-05	0e+00
49	0.001	0.09315	−0.0030	5084	0.0015	1e-16
49	0.01	0.01086	−0.0031	583.2	0.0125	2e-18
49	0.1	0.000964	−0.0029	56.06	0.116	-2e-19

Table S5: Double well, normalized compression loss: near-exact Bayes loss, two-rate approximation [Eq. (13) of the main text], single-rate Gaussian theory, and the loss estimated from the skill difference of a classifier trained on 8.4×10^3 points (`S4e_two_rate_leak.csv`, `S4_summary.csv`).

τ_c	w	$\bar{\lambda} w$	exact	two-rate	single-rate	classifier-skill estimate
∞	5	0.50	0.061	0.048	0.203	0.061
∞	10	1.00	0.132	0.102	0.403	0.130
∞	20	2.00	0.245	0.181	0.637	0.247
∞	40	4.00	0.394	0.289	0.824	0.432
∞	80	8.00	0.549	0.433	0.921	0.706
80	5	0.52	0.051	0.074	0.210	0.058
80	10	1.04	0.110	0.143	0.413	0.137
80	20	2.08	0.206	0.226	0.648	0.262
80	40	4.16	0.332	0.310	0.831	0.415
80	80	8.32	0.479	0.403	0.924	0.719
20	5	0.51	0.054	0.074	0.206	0.051
20	10	1.02	0.116	0.144	0.408	0.103
20	20	2.04	0.217	0.227	0.642	0.185
20	40	4.07	0.350	0.314	0.827	0.308
20	80	8.15	0.498	0.411	0.923	0.502
5	5	0.51	0.058	0.057	0.207	0.065
5	10	1.02	0.124	0.115	0.408	0.147
5	20	2.04	0.231	0.191	0.643	0.297
5	40	4.08	0.371	0.282	0.828	0.520
5	80	8.16	0.531	0.396	0.923	0.843

Table S6: Entropy production of the PCA(2) states against reversible nulls: amplitude-adjusted surrogates (200 per market; literal surrogates, 100), symmetric-sign surrogates and random signs on the actual $|r_t|$; null max is the largest value in the null ensemble (`E4_summary.csv`, `E4_irreversibility_surrogates.csv`, `E4e_symmetric_sign.csv`, `E4c_pca_factorization.csv`).

state	null	observed $\dot{e}_p$	null median [5%, 95%]	null max
BTC	reversible surrogate	0.0014	0.0029 [0.0002, 0.0079]	0.0120
ETH	reversible surrogate	0.0016	0.0026 [0.0000, 0.0095]	0.0136
SPX	reversible surrogate	0.0352	0.0027 [0.0004, 0.0070]	0.0117
SPX	literal iaaft	0.0352	0.0028 [0.0005, 0.0075]	0.0115
SPX	symmetric-sign surrogates	0.0352	0.0029 [0.0003, 0.0080]	0.0135
SPX	random signs on actual $\|r\|$	0.0352	0.0029 [0.0000, 0.0103]	0.0121

Table S7: Stochastic-volatility null, all variants: G1/T1 Gaussian/Student-t shocks with one volatility factor, G2/T2 with two factors; 30 replicates per cell (`E4f_sv_variants.csv`, calibration in `E4f_calibration.csv`).

variant	ρ	$\dot{e}_p$ median [5%, 95%]	ω_{win}	ω_{lev}	L_+
G1	0.00	0.0020 [0.0003, 0.0059]	−0.0200	+0.0005	-11
G1	-0.40	0.0057 [0.0008, 0.0113]	−0.0201	+0.0123	-109
G1	-0.60	0.0193 [0.0122, 0.0279]	−0.0201	+0.0179	-207
G1	-0.75	0.0362 [0.0218, 0.0489]	−0.0203	+0.0243	-222
G1	-0.85	0.0440 [0.0279, 0.0604]	−0.0196	+0.0268	-260
G1	-0.95	0.0564 [0.0317, 0.0752]	−0.0201	+0.0301	-295
T1	0.00	0.0020 [0.0003, 0.0073]	−0.0234	+0.0003	-2
T1	-0.40	0.0035 [0.0005, 0.0078]	−0.0234	+0.0111	-112
T1	-0.60	0.0107 [0.0060, 0.0209]	−0.0222	+0.0166	-197
T1	-0.75	0.0203 [0.0119, 0.0303]	−0.0234	+0.0202	-236
T1	-0.85	0.0329 [0.0144, 0.0512]	−0.0237	+0.0248	-280
T1	-0.95	0.0321 [0.0002, 0.0597]	−0.0230	+0.0270	-323
G2	0.00	0.0031 [0.0007, 0.0076]	−0.0249	+0.0004	-5
G2	-0.40	0.0091 [0.0022, 0.0187]	−0.0252	+0.0154	-176
G2	-0.60	0.0223 [0.0000, 0.0442]	−0.0245	+0.0241	-266
G2	-0.75	0.0227 [0.0028, 0.0594]	−0.0254	+0.0281	-308
G2	-0.85	0.0359 [0.0019, 0.0756]	−0.0260	+0.0314	-319
G2	-0.95	0.0369 [0.0025, 0.0797]	−0.0240	+0.0333	-393
T2	0.00	0.0031 [0.0007, 0.0090]	−0.0293	0.0000	9
T2	-0.40	0.0044 [0.0002, 0.0135]	−0.0278	+0.0139	-176
T2	-0.60	0.0134 [0.0011, 0.0290]	−0.0278	+0.0224	-242
T2	-0.75	0.0212 [0.0009, 0.0524]	−0.0280	+0.0255	-346
T2	-0.85	0.0231 [0.0050, 0.0520]	−0.0276	+0.0287	-337
T2	-0.95	0.0188 [0.0031, 0.0372]	−0.0276	+0.0312	-363

Table S8: Third-order time asymmetry of $|r_t|$ against 200 amplitude-adjusted surrogates (`E4e_third_order.csv`).

statistic	observed	reversed	IAAFT null [2.5%, 97.5%]	p
$\sum_{k=1}^{20}\langle z_t^2 z_{t+k} - z_t z_{t+k}^2\rangle$, $z = \|r\|$	1.026	-1.026	0.019 [-0.707, 0.842]	0.010

Table S9: Entropy production per step of the OU model fitted to two trailing windows $(w, 4w)$ of x in the double well, the Gaussian-observer prediction with $\bar{\lambda}$, and the latent (x, c) value from the same estimator (`S5_summary.csv`; script `S5_doublewell_twowindow.py`).

τ_c	w	$\dot{e}_p$ observed	Prop. 2	latent (x, c)	R observed
20	1	7.643	7.071	0.0031	1.782
20	5	4.002	2.553	0.0031	1.160
20	10	3.711	1.470	0.0031	1.076
20	20	3.460	0.713	0.0031	1.036
20	40	2.847	0.306	0.0031	1.018
20	80	1.854	0.130	0.0031	1.010
5	1	7.686	7.082	0.0029	1.781
5	5	4.054	2.565	0.0029	1.158
5	10	3.763	1.480	0.0029	1.075
5	20	3.455	0.719	0.0029	1.035
5	40	2.807	0.309	0.0029	1.017
5	80	1.757	0.132	0.0029	1.010
80	1	7.590	7.073	0.0037	1.783
80	5	3.922	2.555	0.0037	1.164
80	10	3.591	1.472	0.0037	1.079
80	20	3.375	0.714	0.0037	1.037
80	40	2.752	0.306	0.0037	1.019
80	80	1.929	0.131	0.0037	1.010
∞	1	7.724	7.072	0.0000	1.778
∞	5	4.023	2.554	0.0000	1.157
∞	10	3.654	1.471	0.0000	1.076
∞	20	3.254	0.714	0.0000	1.037
∞	40	2.528	0.306	0.0000	1.019
∞	80	1.534	0.131	0.0000	1.011

Table S10: Moving-block-bootstrap standard errors of the entries of Table II, in units of 10^{-4}, for blocks of 20 / 60 days (1000 replicates; forward, time-reversed and centered series aligned on the original time axis and resampled jointly; means and variances recomputed in every replicate). Pairs: lag-one pairs common to the three series (`E4g_bootstrap_se.csv`; script `E4g_bootstrap_sensitivity.py`, seeds [20261028, k, L]).

sample	channel	pairs	SE(ω)	SE(ω^{rev})	SE(ω^{cen})	SE(O)	SE(P)
S&P 500	window	6239	19 / 24	21 / 25	14 / 12	17 / 24	11 / 7
S&P 500	leverage, return	6239	34 / 29	35 / 30	34 / 29	28 / 14	21 / 26
S&P 500	leverage, drawdown	6239	30 / 27	28 / 20	32 / 22	22 / 14	18 / 19
S&P 500	PCA(2) state	6239	32 / 27	29 / 30	30 / 26	25 / 18	18 / 22
S&P, to 2020	window	4606	22 / 28	23 / 30	16 / 12	20 / 28	12 / 9
S&P, to 2020	leverage, return	4606	37 / 32	37 / 33	37 / 32	28 / 14	24 / 29
S&P, to 2020	PCA(2) state	4606	29 / 27	30 / 29	30 / 27	22 / 18	19 / 22
Bitcoin	window	4392	28 / 30	27 / 28	26 / 22	22 / 27	16 / 12
Bitcoin	leverage, return	4392	41 / 43	42 / 43	41 / 44	29 / 18	30 / 39
Bitcoin	PCA(2) state	4392	34 / 37	46 / 47	40 / 36	28 / 19	30 / 37
Ether	window	3237	29 / 33	35 / 34	33 / 23	26 / 31	19 / 12
Ether	leverage, return	3237	36 / 39	36 / 39	39 / 41	24 / 15	27 / 36
Ether	PCA(2) state	3237	31 / 31	43 / 42	43 / 36	25 / 17	28 / 33

Table S11: What the market's windows cost in information: normalized compression loss ℓ of the event that the latent log variance of the one-factor stochastic-volatility model ($\phi = 0.98$, Appendix E of the main text) exceeds 2σ within ten days. Lemma 1 for noise-free trailing windows of the latent state; simulation (10^6 days) for the observers of the market pipeline and, as a check, for the noise-free pair, with the loss estimated as the out-of-sample mean squared error of a gradient-boosted regression of the oracle on the observation (`E9_market_information.csv`; script `E9_market_information.py`, seed 20261029).

observer	method	λw	ℓ
window 5	Lemma 1 (noise-free windows of x)	0.10	0.109
window 20	Lemma 1 (noise-free windows of x)	0.40	0.396
window 60	Lemma 1 (noise-free windows of x)	1.21	0.719
window 250	Lemma 1 (noise-free windows of x)	5.05	0.952
windows (5,20)	Lemma 1 (noise-free windows of x)	0.40	0.102
windows (5,20)	simulation, regression		0.105
log RV (5, 20 days)	simulation, regression		0.448
PCA(2) state	simulation, regression		0.493

Table S12: Data files behind the figures and tables of the main text.

item	files in `reproducibility_pre/tables_pre/` (unless stated)
Fig. 1	data/spx_2000_2026_r1.csv
Fig. 2	S1_gaussian_leak.csv, T2c_observer_curves.csv, S2_gaussian_irrev.csv
Fig. 3	S4d_acf.csv, S4e_two_rate_leak.csv, S5_summary.csv
Fig. 4	E4e_pairs.csv, E4c_pca_factorization.csv, E4e_surrogate_bands.csv, E4e_symmetric_sign.csv, E4f_sv_variants.csv
Table I	T2d_table1.csv (script T2d_table1_horizon.py)
Table II	E4e_pairs.csv, E4c_pca_factorization.csv, E4e_surrogate_bands.csv, E4g_bootstrap_se.csv
Table III	E4f_sv_variants.csv, E4c_pca_factorization.csv, E4e_pairs.csv
numbers in the text	numbers.tex, written by P1_numbers.py with sources in P1_numbers_sources.csv